Few-shot fatigue limits after a provenance audit

AM316L | Working technical brief | Computational benchmark, not a validated safe-design method

Question

Can a few endpoints from a new AM316L condition make lower-life limits both safer and useful? We tested censored monotone Basquin baselines, target-intercept updating and hierarchical borrowing, using fixed support/query splits and held-out source blocks.

Concrete evidence

Original-methods review reduced a 1,126-point public pilot to 309 points/32 curves/11 method-reviewed source blocks for the main cohort. It identified stepped-load observations, republished curves and source-supported runout/treatment corrections. Physical specimen/batch independence remains incompletely documented. These are corrections to this open-data benchmark, not proof of errors in Centola et al.'s original 518 records.

Results

In the secondary full-training raw track, three endpoints per condition increased geometric lower life 1.92x and reduced mean danger identification endpoints from 13.35-15.41% to 8.31-8.93%. The descriptive 95% source-bootstrap life-gain interval is 0.84-3.65x; risk worsens for several sources. The prespecified 90% calibration becomes a zero-life bound with the available source count, so its usefulness criterion fails.

An acquired author-provided campaign contains 129 complete specimens in three diameter groups, representing one related campaign. Three random endpoints per group (nine total) reduce danger from 53.66% to 13.19% by lowering overly optimistic limits. A matched simple target-intercept control gives 6.99% at shorter life. No hierarchical superiority or general 90% source-conditional guarantee is established. The campaign uses a crack-related ultrasonic endpoint, not necessarily full fracture.

A narrow next step

A useful continuation would test stress-level selection at fixed query sets and matched experimental budgets against a random-fatigue-limit baseline and simple local updating. This needs specimen/build lineage, consistent failure definitions, actual censoring plans and additional held-out campaigns. A source-to-condition mapping or brief methodological feedback would resolve a concrete bottleneck.

Code, split receipts, aggregate figures and independent arithmetic checks are available locally. No new physical experiment or correspondence was performed. Original data DOI: 10.6084/m9.figshare.22337629.v2 (CC BY 4.0). External article: 10.1038/s41598-025-97031-1; repository data license unspecified, so raw external records are excluded from the shareable archive.